

IGDTUW ANVESHAN FOUNDATION

6 WEEKS INTERNSHIP ON MACHINE LEARNING AND GENERATIVE AI USING PYTHON

College: Indira Gandhi Delhi Technical University for Women

Program: B. TECH

Department: Computer Science and Engineering

Enrollment Number: 23201012025

Student Name: Yumna Nasir

ASSIGNMENT 3

INPUT:

```
Assignment_3_Yumna_Nasir_23201012025.py - C:/Users/Yumna Nasir/OneDrive/Desktop/INTERN/Assignment_3_Yumna_Nasir_23201012025.py (3.14.2)
File Edit Format Run Options Window Help

#ASSIGNMENT 3

#1. Create a function to print first 10 natural numbers.
def printNaturalNumbers(n):
    for i in range(1, n + 1):
        print(i, end=" ")
    print()
if __name__ == "__main__":
    n = 10
    printNaturalNumbers(n)

#2. Create a function to calculate sum of first N natural numbers.
def findSum(n):
    sum = 0
    i = 1
    # Iterating over all the numbers between 1 to n
    while i <= n:
        sum = sum + i
        i = i + 1
    return sum
if __name__ == "__main__":
    n = 5
    print(findSum(n))

#3. Create a function to reverse a number.
def rev_num(n):
    if n < 10:
        return n
    return int(str(n % 10) + str(rev_num(n // 10)))
n = 987654
print(rev_num(n))

#4. Create a function to count digits in a number
def countDigit(n):
    if n == 0:
        return 1
    count = 0
    while n != 0:
        n = n // 10
        count += 1
    return count
if __name__ == "__main__":
    n = 58964
    print(countDigit(n))
```

```

File Edit Format Run Options Window Help

#5. Create a function to check palindrome number.
import math
def rev(num):
    return int(num % 10) and (num % 10) * \
        (10**int(math.log(num, 10))) + \
        rev(num // 10)

test_number = 9669669
print("The original number is : " + str(test_number))
res = test_number == rev(test_number)
print("Is the number palindrome ? : " + str(res))

#6. Create a function to generate Fibonacci series.
def fib(n):
    if n <= 1:
        return n
    return fib(n - 1) + fib(n - 2)
for i in range(7):
    print(fib(i), end=" ")

#7. Calculator Using Functions that contains the following features:
def add(n1, n2):
    return n1 + n2
def sub(n1, n2):
    return n1 - n2
def mul(n1, n2):
    return n1 * n2
def div(n1, n2):
    return n1 / n2
print("Please select operation -\n"
      "1. Add\n"
      "2. Subtract\n"
      "3. Multiply\n"
      "4. Divide\n")
sel = int(input("Select operation (1-4): "))
n1 = int(input("Enter first number: "))
n2 = int(input("Enter second number: "))

if sel == 1:
    print(n1, "+", n2, "=", add(n1, n2))
elif sel == 2:
    print(n1, "-", n2, "=", sub(n1, n2))
elif sel == 3:
    print(n1, "*", n2, "=", mul(n1, n2))
elif sel == 4:
    print(n1, "/", n2, "=", div(n1, n2))
else:
    print("Invalid input")

```

Ln: 110 Col: 13

```

File Edit Format Run Options Window Help

#8. Create a text file and store student details.
def write_student_data(num_students):
    with open("pythonml.txt", "w") as file:
        for i in range(num_students):
            roll_number = int(input("Enter roll number for student {}: ".format(i+1)))
            name = input("Enter student {}'s name: ".format(i+1))
            marks = float(input("Enter student {}'s marks: ".format(i+1)))
            student_data = "Roll Number: {}, Name: {}, Marks: {}".format(roll_number, name, marks)
            file.write(student_data)
num_students = int(input("Enter the number of students (maximum 3): "))
if num_students > 3:
    print("Error: Maximum of 3 students allowed.")
else:
    write_student_data(num_students)
    print("Student data written to student.txt successfully!")

#9. Read data from a file.
file = open("pythonml.txt", "r")
content = file.read()
print(content)
file.close()

#10. Handle division by zero using exception handling.
a = 16.0
b = 0
try:
    res = a / b
except ZeroDivisionError:
    res = 0
print(res)

```

#11. Create a Student class with name and marks.

```

class Student():
    id = 0
    name = ""
    gender = ""
    total = ""
    per = 0
    def setData(self, id, name, gender, total, per):
        self.id = id
        self.name = name
        self.gender = gender
        self.total = total
        self.per = per
    def showData(self):
        print("Id :", self.id)
        print("Name :", self.name)
        print("Gender :", self.gender)
        print("Total :", self.total)
        print("Percentage :", self.per)
s = Student()
s.setData(1, 'Sam Kumar', 'Male', 422, 84.44)
s.showData()

```

Ln: 90 Col: 5

OUTPUT:

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Yumna Nasir/OneDrive/Desktop/INTERN/Assignment_3_Yumna_Nasir_23201012025.py
1 2 3 4 5 6 7 8 9 10 15
456789
5
The original number is : 9669669
Is the number palindrome ? : True
0 1 1 2 3 5 8 Please select operation -
1. Add
2. Subtract
3. Multiply
4. Divide
Select operation (1-4): 3
Enter first number: 9
Enter second number: 10
9 * 10 = 90
Enter the number of students (maximum 3): 2
Enter roll number for student 1: 12
Enter student 1's name: Ayra
Enter student 1's marks: 89
Enter roll number for student 2: 33
Enter student 2's name: Inaya
Enter student 2's marks: 56
Student data written to student.txt successfully!
Roll Number: 12, Name: Ayra, Marks: 89.0Roll Number: 33, Name: Inaya, Marks: 56.0
0
Id : 1
Name : Sam Kumar
Gender : Male
Total : 422
Percentage : 84.44
```